Project Overview

WATCHtower (Worldwide Analysis of Threats, Conflicts & Hazards) is a geopolitical intelligence and visualization platform designed to aggregate, organize, and present information concerning international developments and state-level conditions.

The platform will integrate information from established governmental and international data sources into a centralized application. Through interactive geographic visualizations, country profiles, historical event records, and risk indicators, WATCHtower aims to make geopolitical information more accessible and easier to interpret.

The primary purpose of WATCHtower is to help users understand current geopolitical conditions, monitor significant international developments, and examine how state-level conditions evolve over time.

Problem Statement

Geopolitical information is distributed across a wide variety of sources such as governmental reports, non-governmental organizations, independent journalists, research institutions, and event databases. As a result of the scatter of information, individuals interested in geopolitics are often required to consult several platforms in order to develop a comprehensive understanding of the geopolitical standing of a particular state. WATCHtower seeks to create a centralized platform to provide users with a consolidated set of relevant information from an accessible interface.

Project Objectives

Objective 1 - Data Aggregation:

Develop a system capable of retrieving, processing, and standardizing geopolitical data from established governmental, international, and research-based sources, including the IMF, World Bank, United Nations, and geopolitical event databases.

Objective 2 - Geographic Visualization:

Create an interactive world map that allows users to navigate between states, visualize geopolitical developments, and access state-specific information through a geographic interface.

Objective 3 - State-Level Analytics:

Create a state-specific profile containing basic information, geopolitical status, travel advisories, relevant news, and other information.

Objective 4 - Comprehensive Risk Assessment:

Develop a live risk assessment methodology to track six significant risk factors.

1. Political - Government stability, corruption, institutional effectiveness
2. Economic - Inflation, debt, unemployment, economic vulnerability
3. Conflict - Armed conflict, casualties, military activity

4. Terrorism - Terrorist incidents, fatalities, threats
5. Crime - Violent crime, organized crime, homicide rates
6. Civil Unrest - Riots, demonstrations, political violence

Each of these six risk factors is used to develop a general risk score out of 100. The purpose of this risk score is to provide users with a general understanding of the overall status of the individual state, not to predict developments that may occur in the future.

Objective 5 - Historical Analysis

Provide a graphical interface displaying how the risk score has developed over time. Each individual risk factor will also have a graphical interface displaying its historical trend.

Objective 6 - Data Transparency:

Ensure that geopolitical information displayed throughout WATCHtower is accompanied by identifiable sources and relevant timestamps, allowing users to independently examine the information underlying the platform's assessments.

Project Scope

The initial development of WATCHtower will focus on creating a centralized platform for monitoring and analyzing geopolitical conditions across countries and territories worldwide.

The system will consist of a backend infrastructure responsible for retrieving, processing, and storing geopolitical information from external sources, alongside a frontend application providing interactive geographic visualizations and state-specific analytical dashboards.

The platform is intended to support the following capabilities:

- Interactive geographic navigation and visualization of geopolitical events.
- State-specific profiles containing relevant geopolitical information, travel advisories, and recent developments.
- Integration of established international datasets and governmental information sources.
- Periodically updated assessments across six geopolitical risk categories.
- Historical tracking and graphical visualization of geopolitical risk indicators.
- Identification of the sources and update dates associated with displayed information.
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Development will occur incrementally, with core geographic visualization, country profiles, and data aggregation prioritized during the initial implementation.

Although WATCHtower will incorporate quantitative risk assessments, it will not initially attempt to predict future conflicts, political crises, or other geopolitical developments. Risk scores will instead provide a simplified representation of current conditions based on available information.

Advanced predictive analytics, machine learning, and additional risk categories may be considered for future development but are outside the initial project scope.